

Altrios from SwRI aims to make the rail freight industry greener

The Southwest Research Institute (SwRI) has launched open source software Altrios, which offers a practical approach to assess green locomotive technologies and infrastructure expansion.

Altrios, a product of collaboration between SwRI, the National Renewable Energy Laboratory, University of Texas at Austin, University of Illinois Urbana-Champaign RailTec, and BNSF Railway, combines locomotive and energy storage tech models with dispatch simulations and routing tools. It scrutinises decarbonisation strategies – like hydrogen, biofuels, and batteries – on real rail infrastructure, aiding informed choices. Altrios facilitates cost-effective evaluation of decarbonisation techniques through computer modelling.

As the US endeavours to decarbonise the transport sector by 2050, Altrios arrives at an opportune moment. Available for public download, Altrios is funded by the US Department of Energy. This software exemplifies the industry's collective efforts to reshape its environmental impact.

In an era of climate consciousness, Altrios will empower rail companies to embrace sustainability without compromising efficiency.



Kandan Kathirvel, chair of the Nephio Technical Steering Committee, said that Nephio R1 focuses on automating cloud infrastructure and network functions using Kubernetes. He underlined the importance of aligning network functions with global infrastructure, a significant move given the shift towards container-based deployment in recent years.

Nephio R1 adopts the Kubernetes Resource Model (KRM) approach for service deployment, replacing the customary Helm Charts. The GitOps-inspired KRM model introduces controlled version management and updates for smoother deployments.



Kathirvel highlighted Nephio's intent-based automation model, stating, "The user has to express the intent for what they really need and then the automation system is capable of identifying what configurations are needed."

Deployed as a separate Kubernetes cluster, Nephio serves as an automation engine for telcos, simplifying workload and cluster automation. Its versatility extends to network functions down to physical switches, with potential for customised Kubernetes operators. The project aims to bolster out-of-the-box automation through additional operators, advancing operational efficiency.

HashiCorp abandons open source for 'source available' approach

HashiCorp has announced a shift from its conventional open source stance to a 'source available' model. This is the company that was behind developing vital software tools such as Vagrant and Terraform. This transition follows the footsteps of MongoDB, Elastic, and Confluent, which have also pivoted to similar strategies, citing concerns about misuse and competition.

Co-founder Arman Dadgar, in a recent blog post, explained that HashiCorp's initial commitment to open source, which prioritised ecosystem building and user



code adaptability, faced challenges due to some entities exploiting open source without contributing back. This prompted HashiCorp's adoption of the Business

Source License (BSL) version 1.1 for future releases, effective from August 10th.

While APIs, SDKs, and various libraries will remain under the Mozilla Public License v2.0 (MPL 2.0), HashiCorp's move to BSL 1.1 means that users can still copy, modify, redistribute, and use the software under specific conditions. However, competitive offerings directly rivalling HashiCorp's products will be restricted.

Avi Press, CEO of Scarf, praised HashiCorp's handling of the transition. He said on Twitter/X, "HashiCorp has set a good bar for how to do a BSL switch smoothly. No misnaming anything, no attacks, just a difficult business decision carefully communicated. They are a well-meaning group of people who have shown they do care about OSS, whether or not you like their decision."

However, Peter Zaitsev, co-founder of Percona, noted the ongoing polarisation within the open source landscape. He described this move as "hostile towards the community who supported the company along the way."

This shift underscores the intricate interplay between community contributions, commercial interests, and competition within the open source domain. As HashiCorp joins the ranks of companies redefining their approaches, the broader open source ecosystem braces for potential impacts.